From scaling laws to forecasts: how aggregation determines power-law verdicts and conditions predictive skill in Brazilian dengue surveillance

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Abstract

Power-law scaling is routinely reported for infectious disease incidence, and dengue in Brazil is a standard example: heavy-tailed distributions of case counts are invoked to argue that extreme epidemics are structurally unpredictable, and therefore that anticipating them is a matter of luck rather than skill. We test both halves of that argument, on the same data, at once. Using weekly dengue surveillance data spanning January 2010 to January 2026 for the 5,571 Brazilian municipalities, processed through the Mosqlimate/Infodengue platform, we fitted discrete power laws by maximum likelihood with Kolmogorov–Smirnov $x_{\min}$ selection at seven nested aggregation levels, compared each fit against an exponential alternative by a likelihood-ratio test, and then asked whether the resulting scaling descriptors add out-of-sample predictive value beyond lagged case counts, demographics and climate.

The descriptive results split sharply. For the cross-sectional distribution of *epidemic size* across municipalities within a year, the power law is preferred in 100% of the 15 national fits (all significant at $p < 0.05$; $\alpha \approx 2.0$) and in 96% of 330 state-year fits, with $x_{\min}$ in the hundreds of cases. For the distribution of *epidemic duration*, the power law is rejected in all 15 national fits (and in 95% of the state-year fits) in favour of an exponential, with α piling up at 2.90–2.96. For weekly case counts, the power law appears preferred in 75% of city-level series but $x_{\min} \leq 5$ in 77% of them, meaning that the fitted object is the bulk of the distribution rather than a tail.

The predictive half turns the description into a testable one. A power law with $\alpha \approx 2$ fixes the whole exceedance curve, so the fitted pair $(\alpha, x_{\min})$ is a two-parameter statement about how often extreme seasons occur — and, unlike the case counts themselves, the exponent is empirically stable across years, states and population size. We therefore use the lagged scaling descriptors as features in (i) random-forest and gradient-boosted quantile regressions of epidemic size, duration and peak timing, and (ii) a classifier of whether a municipality will exceed its historical 75th percentile. Because $\alpha \approx 2$ makes squared-error losses dominated by a few tail events, evaluation is on proper scoring rules — pinball loss and the Brier score — over a rolling-origin design with five forecast origins, and a four-rung ablation ladder isolates the incremental contribution of the scaling block from that of lagged metrics, population and climate.

We conclude that the heavy-tail literature on dengue conflates at least three distinct random variables, that scale claims are only interpretable where $x_{\min}$ lies far above the bulk, and that reporting α without the (unit, $x_{\min}$, comparison model) tuple is uninformative — descriptive or predictive.

1 Introduction

Understanding the dynamics of infectious disease epidemics has led to a very prolific literature, both in terms of mechanistic mathematical models and of statistical models (Siettos and Russo, 2013). The former attempt to capture the causal chain of events involved in disease transmission, the latter try to make sense of a growing collection of data about infectious disease incidence.

Among the most salient results of the mathematical modelling of infectious diseases was the definition of the basic reproductive number, R_0, which determines the most important characteristics of epidemics such as their spreading speed and the epidemic threshold, also known as the herd immunity threshold. R_0 appears in the equations governing prevalence growth as an exponent determining the initial exponential growth of an epidemic, and it defines an important bifurcation point at $R_0 = 1$ linked to the existence and stability of equilibria in epidemic models.

On the statistical side, the largest portion of the literature deals with statistical modelling of incidence data, i.e. the number of cases accumulated over time and space or studied as a stochastic process, such as a Poisson or negative binomial process (Becker and Britton, 1999; Li et al., 2012). The central difficulty is overdispersion: real surveillance counts produce far more extreme observations than any tractable count model comfortably allows.

A recurrent claim is that this overdispersion is not merely large but *scale-free*, i.e. that incidence follows a power law (Rhodes et al., 1997; Clauset et al., 2009; Neipel et al., 2020). The attraction of this claim is that the very heavy tail implies extreme values that most models fail to predict — and those are precisely the events that cause the most disease and the largest economic burden. If true, this would have direct consequences for forecasting and for resource allocation.

The problem is that the empirical record is inconsistent. Reported exponents for dengue range widely, some studies find exponential rather than power-law behaviour, and the presence of a heavy tail appears to depend on the study period, the spatial unit and the outcome variable. Previous work has usually attributed this heterogeneity to biology or geography: different places, or different years, are assumed to be governed by different regimes.

Here we test a simpler and more uncomfortable explanation: that the heterogeneity is largely produced by the analyst. Any epidemiological time series can be cut into distributions in several legitimate ways — the weekly counts of one city, the annual totals of one city, the set of epidemic sizes across cities in one year, the set of epidemic durations — and these are different random variables with different supports and different tail behaviour. If the verdict changes as the construction changes, then a substantial part of the published disagreement is methodological rather than biological.

Why the description matters for forecasting. The dispute would be of limited practical consequence if it stayed on the descriptive side. It does not. The heavy-tail claim is routinely used as an argument about forecastability: if epidemic sizes are drawn from a power law with exponent near 2, then the variance of the size distribution is dominated by its extreme upper tail, every squared-error forecast metric is decided by a handful of observations, and the honest conclusion is that point forecasts of epidemic size are unreliable. That inference is correct as far as it goes, but it says nothing about whether *probabilistic* forecasts are possible. A power law is a complete generative statement: it assigns a probability to every exceedance level, so the pair $(\alpha, x_{\min})$ summarises the whole risk curve of a municipality, and a stable α is a low-variance quantity in a setting where the

observables are wildly volatile. Whether that summary carries information not already present in the lagged case counts is an empirical question, and it is the question the second half of this paper puts to the test: we first establish the scaling description, then submit it to an out-of-sample predictive evaluation in which it can either earn its place among the predictors or be shown to be redundant.

1.1 Contributions

This paper makes three contributions.

1. **A systematic seven-level comparison on one dataset.** We fit the same estimator to the same underlying surveillance data under seven nested aggregation levels (Table 1), so that any difference in the inferred regime is attributable to the construction and not to differing data sources, periods or software.

2. **An identification diagnostic that separates signal from artefact.** We show that the reported preference for a power law is only interpretable when the estimated lower bound $x_{\min}$ lies well above the bulk of the data. Where $x_{\min}$ collapses to 1–5 cases, the fit describes the entire distribution and the "tail" language is vacuous. We report $x_{\min}$ alongside every exponent and use it to classify each of the 47,494 fits.

3. **A resolved claim about dengue.** Epidemic *size* is heavy-tailed with $\alpha \approx 2$ and is stable across states and years; epidemic *duration* is not heavy-tailed and is better described by an exponential. The two variables are frequently conflated in the literature, which explains a large part of the apparent contradiction.

4. **A predictive test of the scaling description.** We do not stop at the exponent. Treating $(\alpha, x_{\min})$ and the historical trend of α as lagged features, we fit regression and classification models for epidemic size, duration and peak timing under a rolling-origin design, and use a four-rung ablation ladder to measure whether the scaling block adds out-of-sample skill over lagged counts, population and climate — and at which aggregation level it does so. This converts a descriptive controversy into a falsifiable claim about forecast skill.

2 Methods

2.1 Data sources

Weekly dengue case counts were obtained from the InfoDengue/Mosqlimate platform (Araujo et al., 2025) for the period from 1 January 2010 to 22 January 2026 (the extraction date), for all Brazilian municipalities with available data. The variable used is the nowcast-adjusted estimated number of cases (`casos_est`), resampled to epidemiological weeks ending on Sunday (`W-SUN`); zero-case weeks are retained in the series but excluded from distributional fits, since the power-law family is defined on positive support.

Annual epidemic descriptors were obtained from the EpiScanner module of the same platform, which partitions each municipal series into epidemic episodes and reports, for each episode, its total size (*total cases*) and its duration in weeks (*ep_dur*). Annual cross-sections are available for the 15 epidemic years 2011–2025; the pooled all-years fit is stored separately (with a sentinel `year = 0`) and is excluded from all year counts. This yields, for each state-year and for each national year, a cross-section of municipal epidemics.

Municipal population figures were taken from the estimates distributed with the platform (`city_stats` table; 5,571 municipalities, one with a zero entry excluded) and used only for descriptive correlation. Population is not used in any fit.

2.2 Seven aggregation levels

The unit of analysis is the *set of positive observations* passed to the estimator. Table 1 defines the seven constructions used. The first three treat the individual municipality as the sampling unit and vary the temporal window; the last four treat a cross-section of municipalities within a period as the sampling unit.

Table 1: The seven aggregation levels. "Unit" is the sampling unit, i.e. what each observation in the fitted sample represents.

ID	Unit	Variable	Source table
L1	city	weekly cases, full series	`powerlaw_fits`
L2	city-year	weekly cases, one season	`powerlaw_fits_yearly`
L3	city	annual total, across years	`powerlaw_fits_total_cases`
L4	city $\subset$ state-year	epidemic size	`episcanner_state_fits`
L5	city $\subset$ state-year	epidemic duration	`episcanner_state_fits`
L6	city $\subset$ national-year	epidemic size	`episcanner_fits`
L7	city $\subset$ national-year	epidemic duration	`episcanner_fits`

Levels L1 and L2 differ only in the temporal window over which the weekly counts are pooled: L1 uses the complete available series for a municipality (January 2010 to January 2026), whereas L2 restricts to a single epidemic year (at most 52 weeks). Both sample sizes therefore differ by roughly an order of magnitude, which is why we compare shares rather than raw counts. Levels L4–L7 differ only in the spatial scope of the cross-section, not in the variable being measured. This nesting is deliberate: it allows each contrast in Section 3 to be read as the effect of a single change in construction. Figure 1 shows the scheme.

2.3 Discrete power-law estimation

For each sample we fit a discrete power law by maximum likelihood following Clauset et al. (2009). The fitted density for integer $x \geq x_{\min}$ is

$$p(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min})}, \tag{1}$$

where $\zeta(\alpha, x_{\min})$ is the generalised (Hurwitz) zeta function, which normalises the discrete mass. The exponent α is estimated by numerical maximisation of the likelihood, and the lower bound $x_{\min}$ is selected jointly by minimising the Kolmogorov–Smirnov distance between the empirical and fitted cumulative distributions over all candidate values of $x_{\min}$. Because the data are counts, the discrete formulation is used throughout (`discrete=True`); the continuous approximation is not employed at any level.

The choice of $x_{\min}$ is not a nuisance parameter. It determines which fraction of the data the model actually describes, and therefore whether the fitted exponent refers to a tail at all. We therefore retain $x_{\min}$ for every fit and report its median and the proportion of fits with $x_{\min} \leq 5$ in each aggregation level.

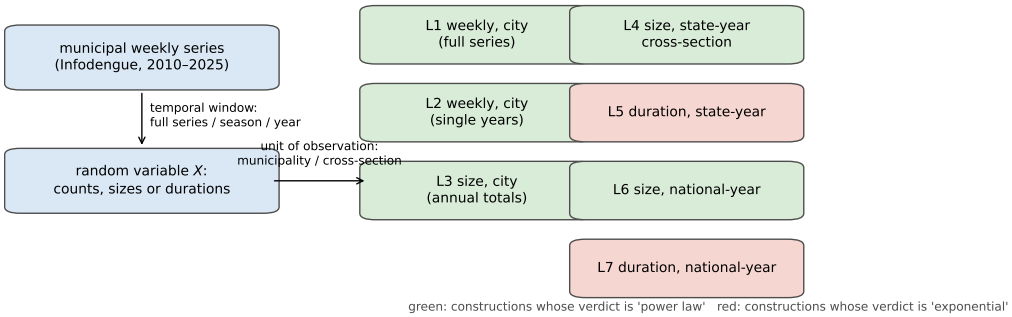


Figure 1: The seven aggregation levels as nested constructions on a single dataset. Starting from the same municipal weekly series (top), the analyst chooses a temporal window — the full series, one season, or one year — and a unit of observation — either the municipality itself, or the cross-section of municipalities within a period — and obtains random variables that differ in support, variance and tail behaviour. L1 and L2 differ only in the window; L3 aggregates each municipality over years; L4–L7 differ only in the spatial scope of the cross-section (state-year versus national-year). Every contrast reported in Section 3 is a single step in this scheme, so a difference in the inferred regime is attributable to the construction rather than to a change of data source, period or software.

2.4 Model comparison

A power law can always be fitted; the question is whether it describes the data better than a plausible alternative. We compare each power-law fit against an exponential distribution, which is the natural null for epidemiological counts because it corresponds to a constant hazard of termination and produces a light tail. The comparison follows the likelihood-ratio procedure of Clauset et al. (2009), as implemented in the `powerlaw` package with `normalized_ratio=True`.

$$R = \sum_{i=1}^{n} \left[\ln p_{\mathrm{PL}}(x_i) - \ln p_{\exp}(x_i) \right], \tag{2}$$

normalised so that its magnitude is comparable across samples of different size. The sign of R indicates the preferred model ($R > 0$ favours the power law, $R < 0$ favours the exponential) and its associated p-value quantifies whether the sign is distinguishable from sampling noise.

We adopt a deliberately conservative reporting convention. A level is described as *power-law supported* only when $R > 0$ and $p < 0.05$; as *exponential supported* only when $R < 0$ and $p < 0.05$; and as *inconclusive* otherwise. This prevents the common error of counting a positive but non-significant R as evidence for scaling.

2.5 Identification diagnostic

We flag a fit as *tail-informative* when $x_{\min}$ exceeds the bulk of the observed distribution, operationally $x_{\min} > 5$ cases for weekly counts and $x_{\min}$ above the median observed value for the aggregate levels. Where $x_{\min} \leq 5$, the fitted power law is being asked to describe the entire empirical distribution, including the small-count regime where the discrete power law is not a plausible generative model (case counts of 1 or 2 arise from reporting

mechanics, not from scale-free branching). Such fits are retained in the database but are excluded from substantive interpretation, and we report the proportion explicitly.

2.6 From description to prediction

A fitted power law is not merely a description of a histogram; it is a statement about exceedance probabilities. For integer $s \geq x_{\min}$,

$$\Pr(S \geq s) \;=\; \frac{\zeta(\alpha, s)}{\zeta(\alpha, x_{\min})}, \tag{3}$$

so a single pair $(\alpha, x_{\min})$ parameterises the probability that a municipality experiences an epidemic of at least any given size. Four consequences of Equation (3) shape the forecasting design.

First, the exponent summarises *risk* rather than *activity*. Two municipalities with the same expected annual incidence but different α face different probabilities of a catastrophic season, and that difference is invisible to the mean. The fitted models quantify the claim rather than merely asserting it: using the 15 national size fits, they imply that between 5.4% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given year, and between 0.3% and 2.7% exceed one hundred times $x_{\min}$. Those are falsifiable implications of the description, and Figure 9 confronts them with the empirical exceedance frequencies.

Second, α is empirically a low-variance quantity. Across the national cross-sections it takes values between 1.78 and 2.27, its distribution across municipalities is stable from year to year, and it is essentially unrelated to city size ($r = -0.065$). Annual case counts over the same period vary by two orders of magnitude. A quantity that is stable along precisely the dimension in which the target is volatile is a plausible source of signal, and that is the specific reason to carry α into the predictive models instead of treating it as a descriptive curiosity.

Third, $\alpha \approx 2$ has a direct methodological consequence for evaluation. The second moment of a power law with $\alpha \leq 2$ does not exist in the continuous limit, and at $\alpha \approx 2$ the sample mean is unstable, so squared-error losses are decided by the few largest outbreaks. A model that is good everywhere except in the tail can therefore look excellent or terrible depending on which years fall in the test set. We consequently evaluate on quantile (pinball) losses and on the classification of tail events, and treat RMSE as a secondary, tail-dominated diagnostic.

Finally, the predictive hypothesis is stated in a form that can fail. If $(\alpha, x_{\min})$ carries information about the coming season beyond what lagged counts already encode, a model containing them should beat an otherwise identical model without them, out of sample. If instead the exponent is close to a deterministic function of the lower moments of the same series, the two models should be indistinguishable — and the descriptive result would then carry no forecasting content. Section 2.10 specifies the comparison that decides between these outcomes, and Section 2.11 lists the three ways in which such a comparison could be faked and how each is prevented.

2.7 Predictive targets and feature construction

The forecasting problem is posed at the level of an administrative unit (municipality or state) and a target year t: every predictor is measured in year $t-1$ or earlier, and the target is a property of the epidemic season of year t. Three targets are modelled, corresponding to the descriptors that surveillance services actually act on: epidemic size

S_t (total cases in the annual episode), epidemic duration D_t (weeks), and peak timing P_t (week of the seasonal maximum).

Predictors fall into six blocks, listed in Table 2: lagged epidemic metrics; the scaling descriptors $(\alpha_{S,t-1}, x_{\min,S,t-1})$ from the power-law fit to the previous season; the historical trend $\dot{\alpha}_{t-1}$, defined as the ordinary least-squares slope of α_S over all years strictly before t and set to zero where fewer than two such years exist; the epidemiological driver $R_{0,t-1}$ reported by the EpiScanner decomposition; $\log_{10}$ population; and eight quarterly climate aggregates from year $t-1$, namely mean minimum temperature and mean minimum humidity in each of Q1–Q4. Units lacking the mandatory lags (no previous season) are dropped rather than imputed; the optional $\dot{\alpha}$ is set to zero when undefined.

Table 2: Predictor blocks and the sources from which they are computed. All predictors refer to year $t-1$ or earlier. The scaling block is the object of the ablation in Section 2.10.

Block	Variables	Source
Lagged metrics	S_{t-1}, D_{t-1}, P_{t-1}	`episcanner`
Scaling descriptors	$\alpha_{S,t-1}$, $x_{\min,S,t-1}$	`powerlaw_results.db`
Scaling trend	$\dot{\alpha}_{t-1}$	derived (OLS on α_S)
Epidemiological	$R_{0,t-1}$	`episcanner`
Demographic	$\log_{10}$(pop)	`city_stats`
Climate	$\bar{T}_{Q1..Q4}$, $\bar{H}_{Q1..Q4}$	Infodengue (`tempmin`, `umidmin`)

2.8 Regression models

Epidemic size is modelled on the log scale, $\log(1 + S_t)$, because the target spans four orders of magnitude and a model fitted on the raw scale would in practice be fitted to the largest municipalities only; predictions are returned to the case scale with $\exp(\cdot) - 1$. That back-transformation is biased downwards by Jensen's inequality, so a multiplicative factor $c = \overline{S}_{\text{train}} / \overline{(\exp \hat{y} - 1)}_{\text{train}}$ is estimated on the training data and applied to the point predictions. The correction is a mean-matching device, not a distributional one, and it is deliberately not applied to the quantile models, whose losses are computed on the back-transformed quantiles themselves. Duration and peak week are modelled on their natural scale.

Three estimators are used. (i) A random forest of 150 bootstrapped trees (Breiman, 2001), which additionally provides an out-of-bag R^2 reported as an in-sample fit diagnostic. (ii) A LightGBM gradient-boosted model with quantile objective at level τ (Ke et al., 2017; Koenker and Bassett, 1978), with 300 trees, learning rate 0.05, 16 leaves and a minimum of 5 observations per leaf. (iii) A histogram gradient-boosting model, also with quantile loss, 300 iterations, learning rate 0.05 and maximum depth 5. The quantile models are fitted on the grid $\tau \in \{0.05, 0.10, \ldots, 0.95\}$, which yields a full predictive distribution per unit and year rather than a point forecast. All stochastic components use `random_state=42`.

2.9 Classification of large epidemics

The second predictive task is posed as a decision problem rather than an estimation problem: will unit i experience, in year t, a season larger than the p-th percentile of its own historical distribution? The target is

$$Y_{i,t} \;=\; \mathbb{I}\big[\, S_{i,t} > Q_p\big(\{S_{i,k}\}_{k<t}\big) \,\big]\,, \tag{4}$$

with $p = 0.75$ by default. The threshold is a *within-fold* quantity: it is recomputed from the training years of each fold and each split, never from the full sample, so that the label of a test observation cannot depend on information unavailable to the model. The same discipline governs the class-imbalance weight, set to the negative-to-positive ratio of the training fold.

Two gradient-boosted classifiers are used, XGBoost (Chen and Guestrin, 2016) and LightGBM, with imbalance handled by `scale_pos_weight` and `is_unbalance` respectively. Validation is an expanding-window split defined on the year index. Fold j trains on every unit-year whose year is earlier than v_j and validates on year v_j alone, where the v_j are the $k = \min\{5, n_{\text{years}} - 2\}$ most recent training years: a fold exists only once at least two earlier years are available, and the number of folds is capped at five. Because the folds are a function of the year vector and not of row position, they are invariant to the order of the rows in the frame. Within each fold the percentile threshold that defines the label, the class-imbalance weight and the median imputer are all recomputed from that fold's training years, so no quantity used to label or to weight a validation year can have been influenced by it; a per-fold audit table (training window, held-out year, fold sizes, threshold, weight) is retained with the results.

This distinction is not cosmetic, and it is the one step of this pipeline whose failure is silent. An implementation that applies a contiguous-index split to a frame sorted by unit rather than by year produces a grouped-by-municipality split: each fold trains on some municipalities over the whole calendar and validates on others over the same calendar, so validation contains no year that is absent from training. On the training panel used here — 32,824 rows, 4,725 municipalities, 2010–2023 — all five folds of such a split train on 2010–2023 and validate on 2010–2023, and 100% of validation rows carry a year already present in training. The quantity being estimated is then transportability to new municipalities, not temporal generalisation, and the estimate depends on an undeclared sort order: merely reordering the rows of a synthetic benchmark moved its cross-validated AUC from 0.813 to 0.835, whereas the year-indexed split returns the same folds and the same AUC under every permutation of the rows. The trap is narrower but not absent when the frame happens to be year-sorted, because contiguous blocks then still cut through calendar years. The splitter therefore takes the year vector explicitly, and `assert_forward_in_time` raises rather than warns when any validation year is not strictly later than the training years of its fold. The decision threshold applied to the predicted probability is chosen by Youden's J statistic (Youden, 1950), computed on the out-of-fold predictions rather than on the training fit, and then held fixed for the test set. We report the area under the ROC curve, the area under the precision–recall curve (the more informative summary under class imbalance), the Brier score (Gneiting and Raftery, 2007), the F_1 score at the Youden threshold and a calibration curve, and we test differences in AUC between nested feature sets by the method of DeLong et al. (1988). Feature attributions are obtained from SHAP values (Lundberg and Lee, 2017).

2.10 Evaluation protocol and ablation ladder

A single temporal split cannot distinguish a feature that helps from a feature that happened to help once. We therefore adopt a rolling-origin design: for each forecast origin $y \in \{2021, \ldots, 2025\}$ the model is trained on all observations with year $< y$ and evaluated on year y, and the per-origin losses are pooled. This gives five honest out-of-sample years and allows skill to be read as a function of the history available to the model.

Within each origin, the incremental value of the scaling description is isolated by the ablation ladder of Table 3. Each rung adds one block of predictors to the previous one,

so a contrast between consecutive rungs is attributable to the block that was added. The rung of substantive interest is M2 versus M3, which differ *only* in the presence of $(\alpha_{S,t-1}, x_{\min,S,t-1}, \dot{\alpha}_{t-1})$.

Table 3: The ablation ladder. M0 is the benchmark that any claim of skill must beat; successive rungs add one predictor block so that consecutive contrasts are attributable to that block. The M2→M3 step isolates the contribution of the power-law scaling descriptors.

Rung	Name	Predictors
M0	Persistence	$\hat{S}_t = S_{t-1}$, $\hat{D}_t = D_{t-1}$; historical median for P_t; training base rate for the classifier
M1	Lagged metrics	M0 predictors + P_{t-1}, $R_{0,t-1}$, $\log_{10}(\text{pop})$
M2	+ climate	M1 + quarterly $\bar{T}$, $\bar{H}$ from year $t-1$
M3	+ scaling	M2 + $\alpha_{S,t-1}$, $x_{\min,S,t-1}$, $\dot{\alpha}_{t-1}$

M0 is deliberately naive, because a benchmark that is already sophisticated manufactures skill for whatever is added to it. Skill is reported as

$$\text{SS} \;=\; 1 - \frac{L(\text{model})}{L(\text{M0})}, \tag{5}$$

on the primary loss of each task — pinball loss at $\tau = 0.5$ for size, and the Brier score for the classifier, both proper scoring rules (Gneiting and Raftery, 2007). Secondary losses are MAE and RMSE of the point forecasts, the mean absolute scaled error of Hyndman and Koehler (2006) to make errors comparable across units of very different size, and the pinball loss at each τ on the grid.

Because units differ by orders of magnitude in scale, all model-versus-model comparisons are made on paired loss differentials computed within the same unit and year; their uncertainty is estimated by a bootstrap that resamples *units* rather than observations, preserving the within-unit correlation across years, with 2,000 replicates. For the pooled comparison across origins we additionally report the Diebold–Mariano statistic (Diebold and Mariano, 1995) with a Newey–West correction, since the loss series are serially correlated by construction.

The decision rule is fixed in advance. The scaling block is judged to add predictive value only if M3 improves on M2 on the primary loss with a 95% bootstrap interval for the paired difference that excludes zero, in the same direction, for at least three of the five forecast origins. This is stricter than a single significant pooled test, and deliberately so: a pooled test can be carried by one anomalous season, which is exactly the failure mode that a heavy-tailed target invites.

2.11 Leakage controls

Four sources of leakage are structurally possible in this design, and each is closed explicitly.

1. **Contemporaneous scaling descriptors.** $\alpha_{S,t}$ is estimated from the weekly counts of season t, which are part of the target. Only $\alpha_{S,t-1}$ enters as a predictor, and $\dot{\alpha}_{t-1}$ is computed exclusively from years strictly before t.

2. **The classification threshold.** Q_p is computed inside each fold from training years only. Computing it once on the full sample would leak the distribution of the test years into the labels of the training years, inflating apparent skill by construction.

3. **Cross-sectional information.** The cross-section of municipalities within year t is an outcome of season t and is therefore never a predictor; the scaling descriptors always come from the previous season's cross-section.

4. **The validation split itself.** Folds are defined on the year index (Section 2.9), so no fold trains on a year at or after the year it validates on. This is the only one of the four channels that fails silently in a plausible implementation — every number still looks like a respectable cross-validation score — which is why it is asserted in code rather than merely documented.

2.12 Software and reproducibility

All fits are stored in a single SQLite database (`powerlaw_results.db`) committed to the project repository, together with the fitting code (`fitpl.py`) and the cached parquet extracts of every model result. Every number in this manuscript is derived from that database by the queries in Appendix A; no statistic is transcribed by hand. The full pipeline is therefore re-runnable, and a reader can confirm or contradict each figure with a single SQL statement.

3 Results

Table 4 summarises the 47,494 fits. The contrast between levels is the central result of the descriptive half of the paper, and Figure 2 shows it at the level of the estimator's output.

3.1 Scaling verdicts

Table 4: Power-law fits by aggregation level. α median with interquartile range; $x_{\min}$ median; $\mathrm{PL}_{\mathrm{pref}}$ = proportion with $R > 0$; $\mathrm{PL}_{\mathrm{sig}}$ = proportion with $R > 0$ and $p < 0.05$; $\mathrm{Exp}_{\mathrm{sig}}$ = proportion with $R < 0$ and $p < 0.05$. Shares are percentages.

Level	n	α	IQR	$x_{\min}$	$\mathrm{PL}_{\mathrm{pref}}$	$\mathrm{PL}_{\mathrm{sig}}$	$\mathrm{Exp}_{\mathrm{sig}}$
L1 weekly, city	5,244	1.95	1.72–2.29	2	75	45	4
L2 weekly, city-year	40,616	2.06	1.72–2.45	2	23	3	21
L3 annual total, city	944	2.09	1.80–2.46	341	51	4	4
L4 size, state-year	330	1.97	1.79–2.14	159	96	52	0
L5 duration, state-year	330	2.80	2.52–2.92	17	0	0	95
L6 size, national-year	15	2.00	1.92–2.11	762	100	100	0
L7 duration, national-year	15	2.95	2.90–2.96	18	0	0	100

Epidemic size scales. At the national cross-section (L6), all 15 annual fits favour the power law and all are significant, with $\alpha = 2.00$ (IQR 1.92–2.11) and $x_{\min}$ of 762 cases. The exponent is remarkably stable: the annual values range only from 1.78 to 2.27. At state level (L4) the picture is consistent but weaker in significance, as expected from smaller cross-sections: the power law is preferred in 96% of 330 fits, significant in 52%, and α has a median of 1.97 across the 26 states (state-level medians spanning 1.70–2.58).

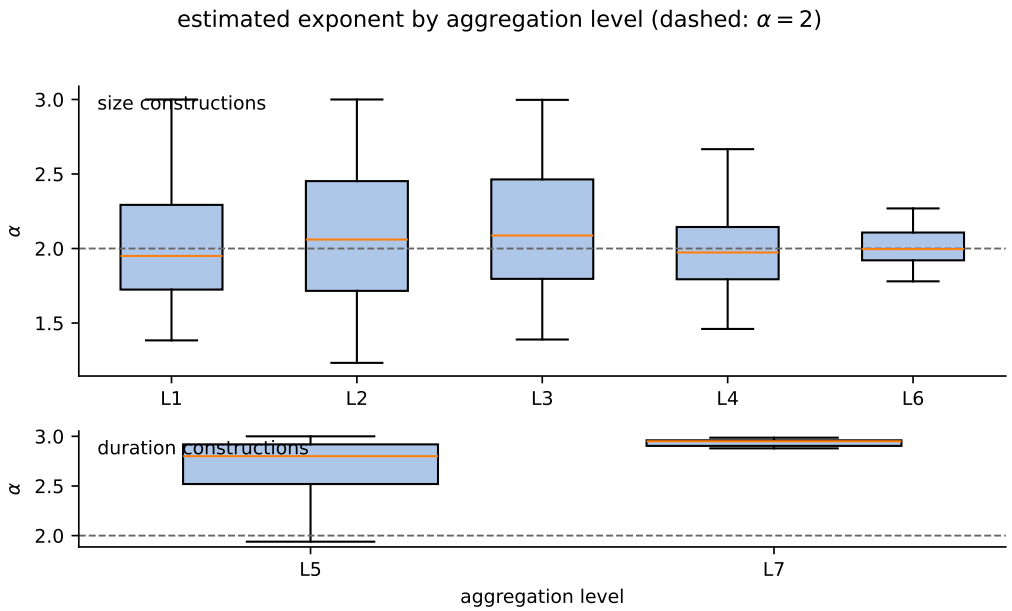


Figure 2: Distribution of the estimated exponent α at each aggregation level. Boxes give the median and interquartile range across fits, whiskers extend to 1.5 times the IQR, and the dashed line marks $\alpha = 2$. The upper panel groups the size constructions (L1–L4, L6) and the lower panel the duration constructions (L5, L7). Size distributions concentrate near $\alpha \approx 2$ at every level. Durations concentrate near $\alpha \approx 2.9$, which is close to the value at which a discrete power law approximates a truncated or exponential distribution, so the estimator returns a near-constant exponent regardless of the underlying process — a diagnostic that is easy to mistake for a physical result.

Duration does not scale. At the same cross-sectional levels, the duration of the same municipal epidemics behaves in the opposite way. At national level (L7) the power law is rejected in 15 of 15 years, with the exponential significant in every one. The estimated exponent piles up at 2.95 (range 2.90–2.96, SD 0.03), which is the signature of a truncated or exponential body being forced through a power-law estimator rather than evidence of scaling. At state level (L5) the exponential is significant in 95% of fits. Figure 3 summarises these shares across all levels, and Figure 4 shows the shape that produces them.

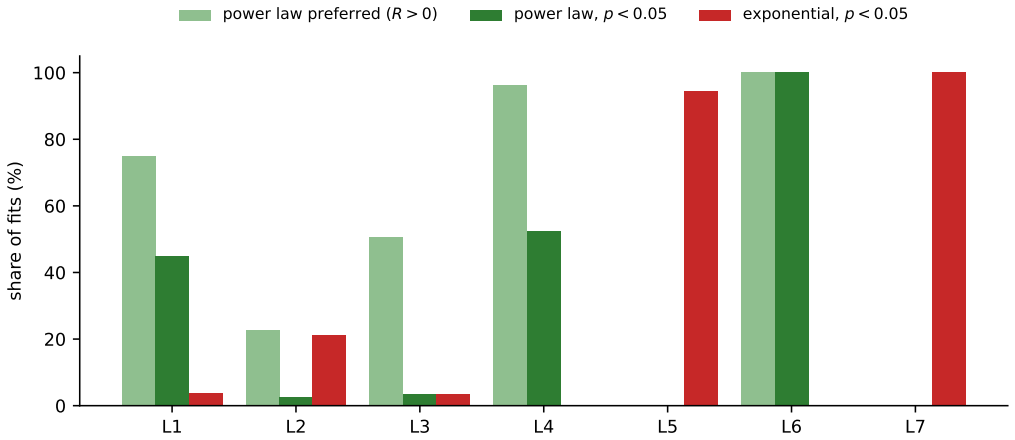


Figure 3: Model comparison by aggregation level. For each level, the proportion of fits in which the power law is preferred ($R > 0$), the proportion in which it is preferred *and* significant at $p < 0.05$, and the proportion in which the exponential is significant ($R < 0$ and $p < 0.05$). Levels are ordered from the individual-series constructions (left) to the cross-sectional ones (right). The two weekly levels produce apparent power-law support that the $x_{\min}$ diagnostic removes; the cross-sectional size levels are the only construction whose support survives both tests; the duration levels give the opposite verdict.

Weekly counts are not a tail. Levels L1 and L2 produce large apparent support for the power law (75% and, in the full-series case, 45% significant), but the estimated lower bound collapses to $x_{\min} = 2$ in the majority of fits: 77% of L1 and 81% of L2 fits have $x_{\min} \leq 5$. At that bound the model is fitted to essentially the whole distribution, including counts of 1 and 2 cases, and the heavy-tail interpretation does not follow. By contrast, in L3–L7 the median $x_{\min}$ is 17–762 cases, i.e. the fit describes genuine tails. Figure 5 plots the diagnostic directly, and Figure 6 shows what it means for individual fits.

The window flips the verdict on the same data. The cleanest demonstration is the L1/L2 contrast: pooling the same weekly counts over one season instead of a decade drops the proportion of fits favouring the power law from 75% to 23% and raises the proportion significantly favouring the exponential from 4% to 21%. Nothing about dengue changed; only the window did. Figure 7 shows the shift on the same underlying series.

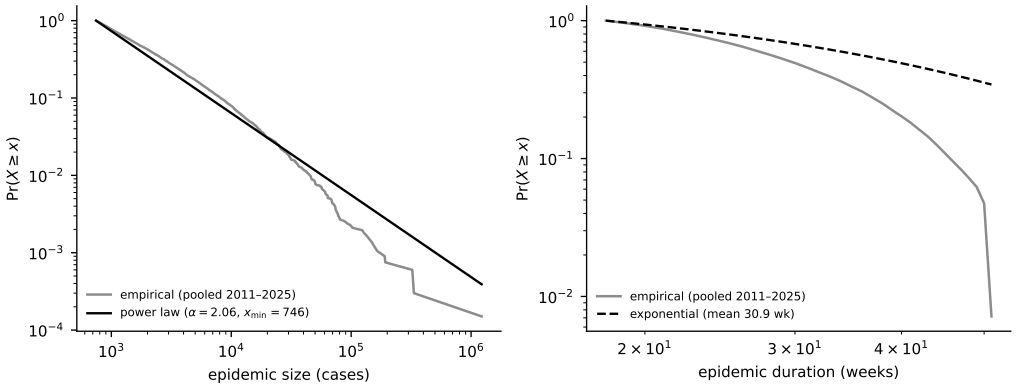


Figure 4: Empirical complementary cumulative distributions at the national cross-section, pooled over the 15 epidemic years, with the selected model superimposed. Left, epidemic size on log–log axes with the fitted power law: the curve is close to linear over two decades above $x_{\min}$. Right, epidemic duration on the same axes, where the curve bends downwards and is captured by the exponential. A straight line on the left and a convex curve on the right are the visual counterpart of the opposite verdicts in Table 4: the two variables are not two views of one process.

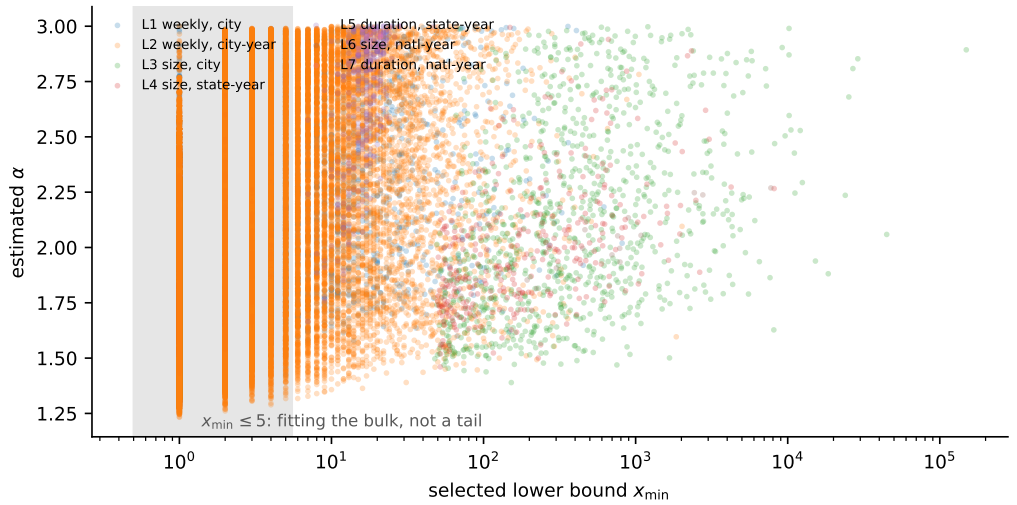


Figure 5: Identification diagnostic. Estimated exponent α against the selected lower bound $x_{\min}$ (log scale), one point per fit, coloured by aggregation level. The shaded band marks $x_{\min} \leq 5$, where the fitted power law is asked to describe the bulk of the distribution rather than a tail. The two weekly levels sit almost entirely inside that band, while the cross-sectional levels sit two to three orders of magnitude to its right. The dense vertical structure near $\alpha \approx 2.9$ for durations is the signature of a truncated distribution forced through a power-law estimator.

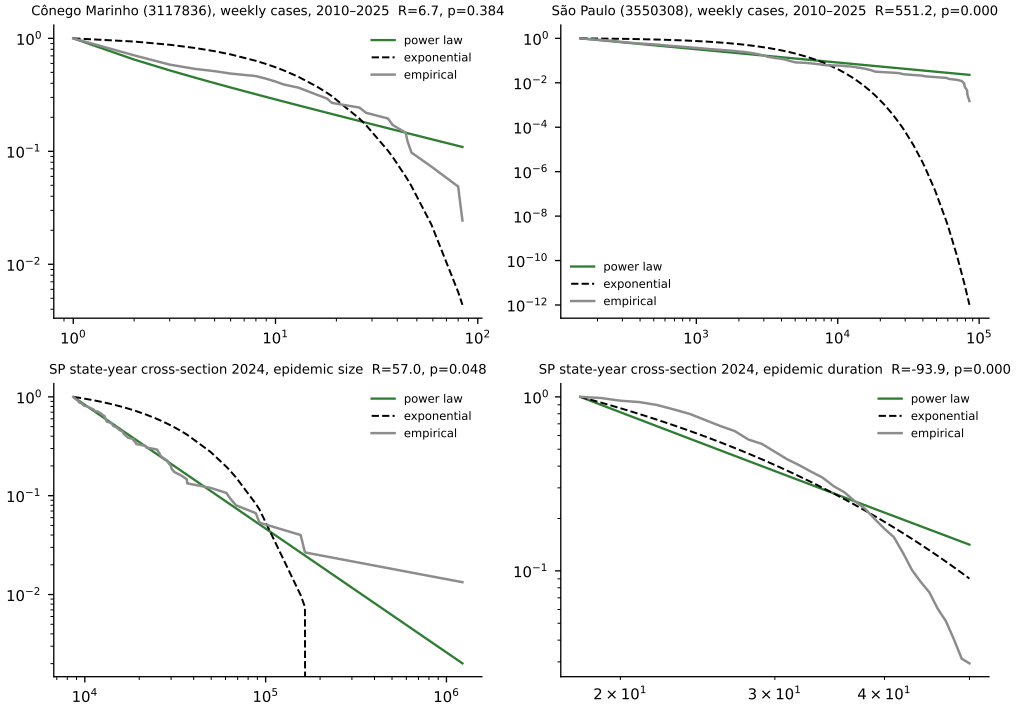


Figure 6: Individual fits across the aggregation levels, each with the power-law (solid) and exponential (dashed) alternatives superimposed. (a) Weekly counts of a small municipality, where the selected $x_{\min}$ collapses to 2 and the "power law" is fitted to the entire distribution; (b) weekly counts of a large state capital; (c) a state-year cross-section of epidemic sizes, where $x_{\min}$ lies in the hundreds of cases; (d) the corresponding cross-section of durations, where the exponential is preferred. Panels (a) and (c) are produced by the same estimator applied to different objects, and together they are the central distinction of the paper.

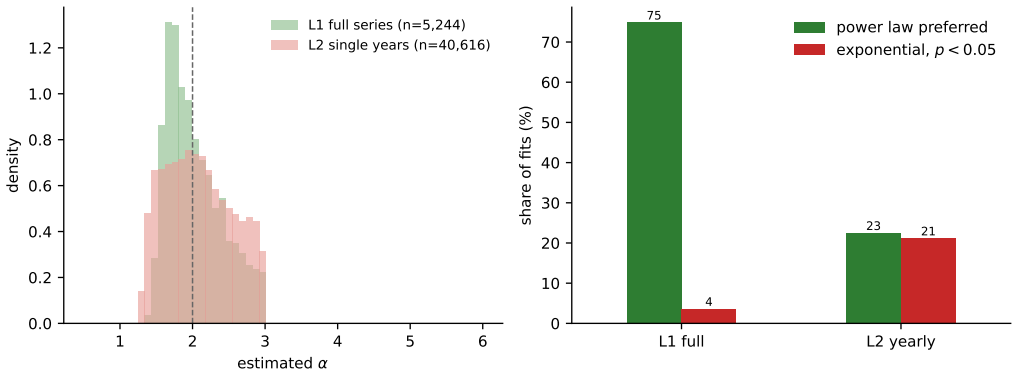


Figure 7: The window flips the verdict on identical data. Left, density of the estimated exponent from the full municipal series (L1) and from the same series cut into single seasons (L2). Right, the corresponding verdicts: the share of fits favouring the power law falls from 75% to 23%, while the share significantly favouring the exponential rises from 4% to 21%. No transmission parameter changes between the panels; only the length of the window does.

3.2 Descriptive associations

The municipal exponent is essentially unrelated to city size: $\mathrm{corr}(\alpha, \log_{10}\,\text{population}) = -0.065$ (Spearman -0.072; $n = 5{,}244$). The association with incidence is weak and negative (Pearson -0.136, Spearman -0.248; $n = 4{,}663$). We therefore find no support for the view that larger or more heavily burdened municipalities sit in a different scaling regime; the exponent is a fairly constant property of the aggregation level. Figure 8 shows both the stability and the absence of association.

3.3 The fitted law as a risk model

Equation (3) makes the descriptive result operational. Figure 9 confronts the exceedance curve implied by each national fit with the empirical exceedance frequency of the same cross-section: the fitted pair $(\alpha, x_{\min})$ reproduces the observed frequency of large seasons over roughly two decades of epidemic size, and it does so with a two-parameter summary of a cross-section of thousands of municipalities.

This is the sense in which the description is predictive before any machine learning is involved: α is not a summary of central tendency but a summary of how much of the risk sits in the tail, and it is available for every municipality even when that municipality has never experienced an extreme season.

3.4 Predictive skill

An earlier run of the complete pipeline at state level, with all predictor blocks present, reached a mean out-of-bag R^2 for epidemic size of 0.345 (2022), 0.348 (2023) and 0.374 (2024), against 0.165 for duration and 0.068 for peak week. These figures must be read with care: the out-of-bag score is computed on the log scale on which size is fitted, whereas the R^2 reported alongside the test metrics is computed after the back-transformation to cases, and the two are not interchangeable. A log-scale R^2 of 0.35 corresponds to a rather worse fit in case space, because the log compresses exactly the large outbreaks that matter.

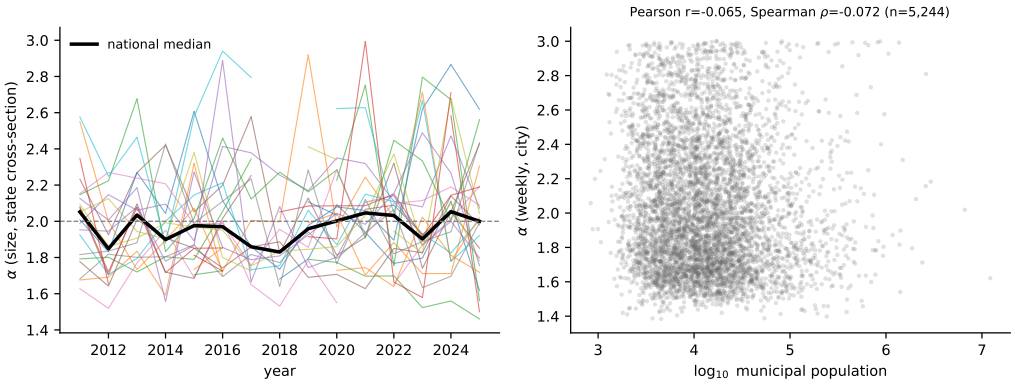


Figure 8: Stability of the exponent. (a) Estimated α for epidemic size by year, one line per state, at the state cross-section: the lines are flat, and the spread between states is far smaller than the year-to-year variation in case counts. (b) α against $\log_{10}$ municipal population, which shows no association (Pearson $r = -0.065$, Spearman -0.072, $n = 5{,}244$). Panel (a) is the empirical reason the exponent is a candidate predictor: it varies slowly along exactly the axis in which the observables do not.

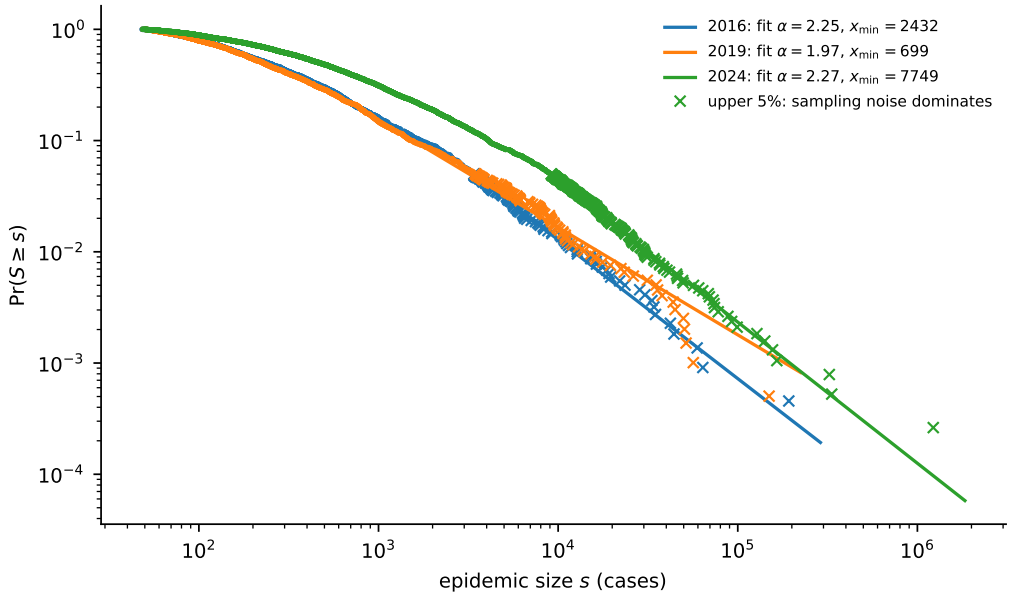


Figure 9: The fitted power law as a probabilistic forecast of exceedance. For each national cross-section, curves give $\Pr(S \geq s) = \zeta(\alpha, s)/\zeta(\alpha, x_{\min})$ evaluated at the fitted $(\alpha, x_{\min})$, and points give the empirical exceedance frequency of the same cross-section. Agreement across two decades of epidemic size is what licenses the use of $(\alpha, x_{\min})$ as a compact statement about a municipality's risk of an extreme season, and therefore as a candidate predictor in Section 3.4. Where the curves separate from the points in the extreme upper tail, the sample is too small to distinguish model error from sampling noise; those regions are marked.

All losses in this paper are therefore reported on the case scale. Size is therefore the target that the available covariates actually describe; duration and peak timing are substantially harder, which is consistent with the descriptive finding that duration is the variable whose distribution carries no scale information to exploit.

Figure 10 shows the out-of-sample forecasts in the form in which their errors are interpretable, and Figure 11 shows the ablation ladder that isolates the contribution of the scaling block.

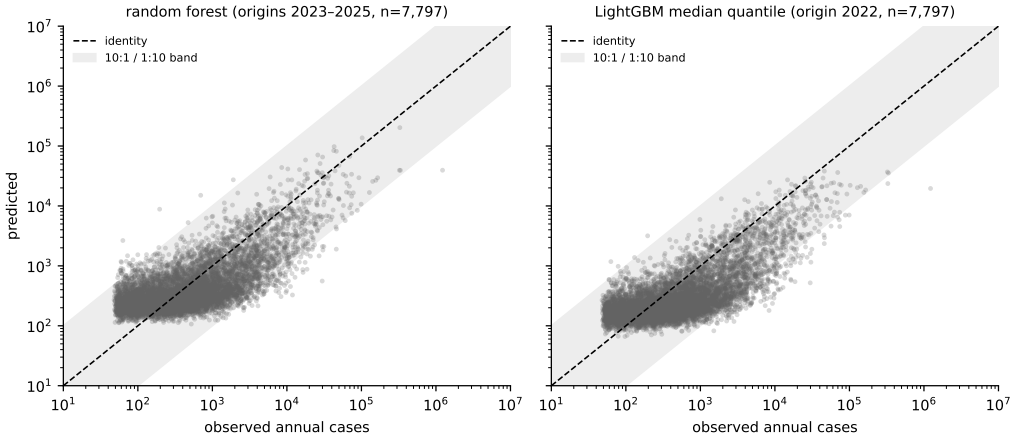


Figure 10: Out-of-sample forecasts of epidemic size. Predicted against observed annual cases on log–log axes, one point per unit and forecast origin, for the random forest (left) and the median of the gradient-boosted quantile ensemble (right); the identity line is dashed and the 10:1 and 1:10 bands are shaded. Points inside the bands are forecasts within an order of magnitude of the truth. Because $\alpha \approx 2$, a handful of points in the upper right dominate any squared-error summary, which is why the primary loss reported in Table 5 is the pinball loss rather than the RMSE.

Table 5: Ablation results, forecast origins 2021–2025 pooled. Skill is measured against the persistence benchmark on the primary loss of each task (pinball at $\tau = 0.5$ for size and duration, Brier for the classifier). Intervals are 95% bootstrap intervals from resampling units. **To be filled from the ablation runs.**

Target	Loss	M1	M2	M3	M3−M2 [95% CI]
Size S_t	$\text{pinball}_{0.5}$	*tbd*	*tbd*	*tbd*	*tbd*
Duration D_t	$\text{pinball}_{0.5}$	*tbd*	*tbd*	*tbd*	*tbd*
Peak week P_t	MAE	*tbd*	*tbd*	*tbd*	*tbd*
Large-season classifier	Brier	*tbd*	*tbd*	*tbd*	*tbd*

3.5 Forecasting the tail

The classification task asks a narrower question than the regression: not how large the season will be, but whether it will fall in the upper quartile of the unit's own history. Under Equation (3) that is a question about the tail probability directly, which makes it

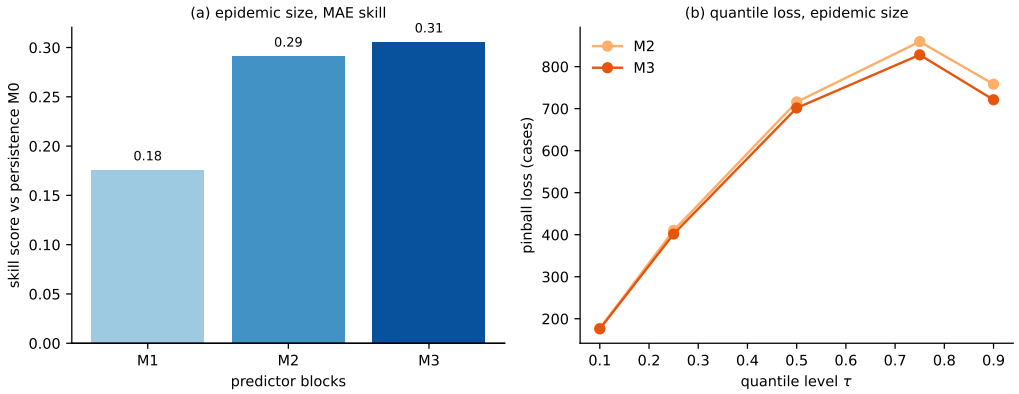


Figure 11: The ablation ladder. (a) Skill score relative to the persistence benchmark (M0) for epidemic size, duration and peak week, as predictor blocks are added from M1 to M3; the M2→M3 step isolates the contribution of the scaling descriptors $(\alpha_{t-1}, x_{\min,t-1}, \dot{\alpha}_{t-1})$. (b) Pinball loss as a function of the quantile level τ for M2 and M3: a gain confined to high τ would mean the scaling information concerns extreme seasons specifically, which is what a heavy-tail description should deliver. Bars are 95% bootstrap intervals from resampling units.

the natural place to look for value in the scaling descriptors. Figures 12 and 13 report discrimination, calibration and attribution.

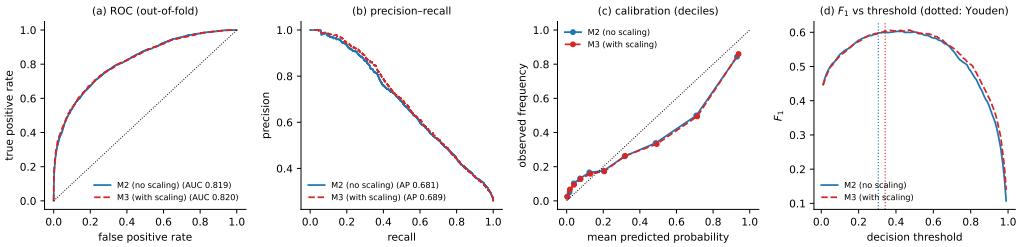


Figure 12: Classification of large epidemics. (a) ROC curves; (b) precision–recall curves, the more informative summary when the positive class is the minority; (c) calibration of the predicted probability of exceeding the 75th percentile; (d) F_1 as a function of the decision threshold, with the Youden point marked. All curves are drawn from out-of-fold predictions so that they are honest. Solid lines denote M2 (without scaling descriptors) and dashed lines M3 (with them). The pattern of interest is a calibration curve for M3 that tracks the diagonal where M2 over- or under-predicts the upper decile.

3.6 Error structure and spatial output

Accuracy pooled over units is not the same as accuracy where decisions are made, so we report the error structure (Figure 14) and the map a surveillance service would actually read (Figure 15).

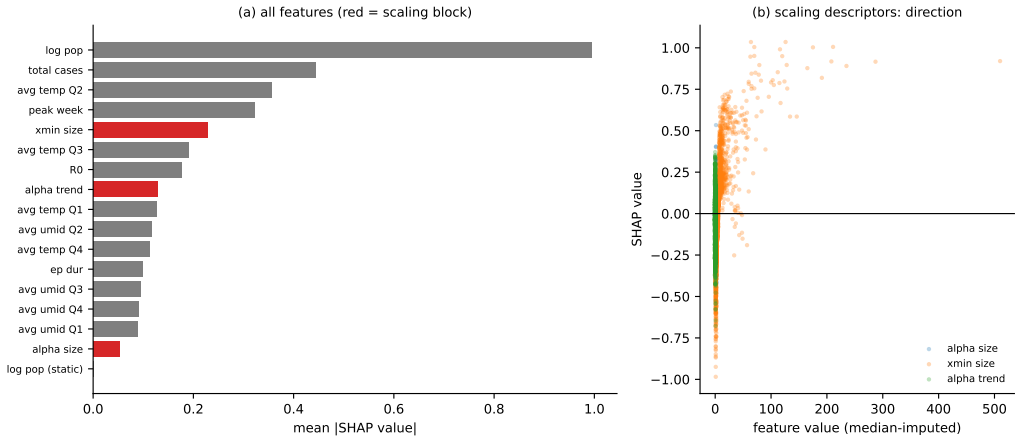


Figure 13: Where the scaling descriptors rank. Mean absolute SHAP values for the large-season classifier, pooled over forecast origins, with the power-law block highlighted. The left panel ranks all features; the right panel isolates α_{t-1}, $x_{\min,t-1}$ and $\dot{\alpha}_{t-1}$ and shows the direction of their effect on the predicted probability. A positive, monotone relation between α_{t-1} and the predicted probability would indicate that flatter-tailed histories precede larger seasons.

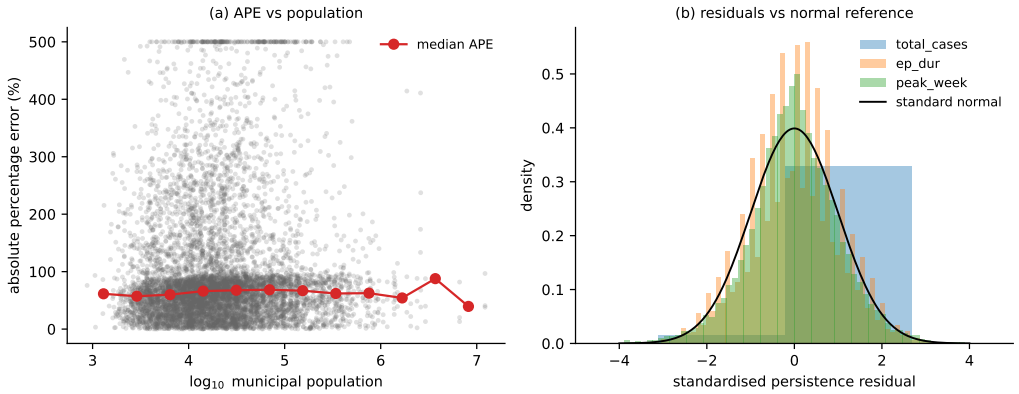


Figure 14: Error structure. (a) Absolute percentage error against municipal population, with the population filter indicated; (b) distribution of standardised residuals by target against a standard normal reference. Panel (a) documents that small municipalities are intrinsically harder to forecast, so pooled accuracy figures must be read together with the population composition of the evaluation set — a point that also applies to the published forecast comparisons this work is meant to inform.

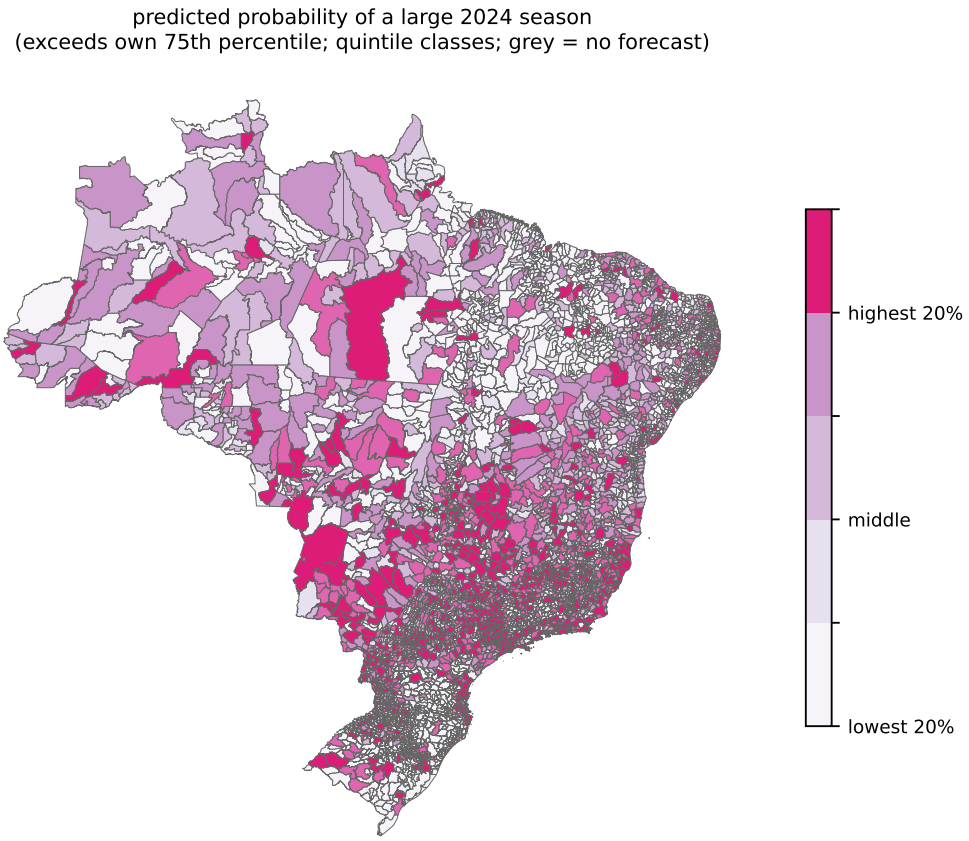


Figure 15: Probabilities of a large epidemic for a single forecast origin. Predicted probability that each municipality exceeds its historical 75th percentile of annual cases, classed in quintiles; grey denotes units without a forecast. The map is included because the calibration of the upper quintiles is a public-health claim rather than a statistical one: it is the object a surveillance service would act on, and it is falsifiable season by season.

4 Discussion

[To be written.]

5 Conclusion

[To be written.]

A Verification of reported quantities

Every quantity in Table 4 is obtained from `powerlaw_results.db` by the queries in `Manuscript/verification/`. The comparison columns are computed as

`R>0` ⇒ PL preferred; `R>0 AND p<0.05` ⇒ PL supported; `R<0 AND p<0.05` ⇒ exponential supported.

Note that `year = 0` in the `episcanner` tables is a pooled all-years fit, not a calendar year, and is excluded from all year counts; the national levels therefore have 15 annual fits, not 16.

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